Question 01 **[50 marks]**

Identify the best solution among these questions and write answers in the provided booklet. **Important: You should need to write down your paper type, which displays in the cover page before you write down the answers.**

(1) Rationalize $\frac{2}{24-\sqrt{214}}$.

a. $\frac{2(24+\sqrt{214})}{-576}$ b. $\frac{2(24-\sqrt{214})}{362}$ c. $\frac{2(24+\sqrt{214})}{362}$ d. $\frac{2(24-\sqrt{214})}{576}$

(2) Let $|A| = 12, |A \cap B| = 5, |B| = 14, |B \cap C| = 7, |A \cup B \cup C| = 100, |A \cap C| = 3\ and\ |C| = 10$. What is the value of $|A \cap B \cap C|$?

a. 49 b. 121 c. None of all d. 79

(3) Let $p: x^2 - 5x + 6 = 0\ where\ x \in \mathbb{Z}$. The statement *p* ?

a. proposition b. Contradiction c. Contingent d. Tautology

(4) The equation of a straight line that passes through the point (3, 4) and perpendicular to the line $3x + 2y + 5 = 0$ is,

a. $2x - 3y + 6 = 0$ b. $2x + 3y + 6 = 0$ c. $2x - 3y - 6 = 0$ d. $2x + 3y - 6 = 0$

(5) In a class there are; 10 students who play cricket and Badminton, 20 students who do not play Badminton or cricket, 13 students who play cricket, 12 students who play Badminton. How many student are there in the class.

a. 25 b. 35 c. 45 d. 55

(6) Let $A = \begin{pmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{pmatrix}$. What is the value of A^2?

a. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ b. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ c. $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ d. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(7) The equation of a line that passes through the points (2, 3) and (1, 5) is,

a. $2x + y - 7 = 0$ b. $2x - y - 7 = 0$ c. $x + 2y - 7 = 0$ d. $2x + y + 7 = 0$

(8) If $A \equiv (6, 4)$ and $B \equiv (2, 12)$ are the two points, then the slope of a line perpendicular to line AB,

a. -2 b. 2 c. 1/2 d. -1/2

(9) The rational and imaginary part of the following $(3i^8 + 15i^{15})(-2i^{37} + i^{48})$ shows in order.

a. -27, -21 b. 33, -9 c. 27, 9 d. None of all

(10) What are the primitive propositions given below.

I. $p_1: 11 - 3 = 8\ and\ 3 + 4 = 8$

II. $p_2: Fish\ can\ fly$

III. $p_3: Fish\ can\ fly\ but\ they\ can't\ swim$

a. I only b. II only c. I and II only d. I and III only

(11) Let p be a true proposition and q be a false proposition. Then,

a. $p \wedge q = T$ b. $q \vee p = F$ c. $p => q = F$ d. $p <=> q = T$

(12) Identify the incorrect statements given below.

I. A relation *R* from an empty set *A* to a non-empty set *B* is a subset of the Cartesian product set $A \times B$.

II. The set of all first elements in a relation *R,* is called the domain of the relation *R*.

III. $(a, b) = (c, d)$ if and only if $a = d$ and $b = c$.

a. I only b. I and II only c. II only d. I and III only

(13) Let $A = \{a, b\}, B = \{1, 2, x\}, and\ C = \{x, y\}$. What is the $|A \times (B \cup C)|$?

a. 2 b. 4 c. 8 d. 10

(14) Identify the real part of the following. $\frac{2i^2+3^2}{4i^3-i^4}$

a. $\frac{17}{7}$ b. $\frac{-7}{17}$ c. $\frac{7}{17}$ d. None of all

(15) Let $p : \mathbb{Z} \subset \mathbb{R}, q : x^2 - 3x + 2 = 0\ when\ x = -1$. Then,

a. $p \wedge q = T$ b. $p \vee q = T$ c. $p => q = T$ d. $p <=> q = T$

(16) Let $C = \{ X \mid X = 5 - n^2\ such\ that\ -2 < n \leq 3\}$. What is the $|P(C)|$, and the largest element of *C*?

a. 32, 14 b. 16, 14 c. 16, 5 d. 64, 5

(17) If the order of matrix A is 4 x 3, the order of B is 4 x 5, and the order of C is 7 x 3, then order of $(A^T B)^T C^T$,

a. 5×3 b. 4×5 c. 5×7 d. 4×3

(18) If $p: a\ number\ from\ \{8, 9, 10, 11, 12\}, q: not\ a\ composite\ number, r: a\ square\ number$, , $s: a\ prime\ number$ then what is the value of $\sim [(p =>\sim q) \wedge (\sim r \vee \sim s)]$?

a. $8, 9, 10, 11, 12$ b. 8, 9, 10 c. 11, 12 d. 11

(19) If $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, then A^4?

a. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ b. $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ c. $\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$ d. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(20) Which is the non-terminating (Recurring) decimal number.

a. π b. $exponential\ value$ c. $4\pi + 2$ d. None of all

(21) Let $A = \begin{pmatrix} x & 2 \\ 4 & 3 \end{pmatrix}$, $A^{-1} = \begin{pmatrix} \frac{1}{8} & \frac{-1}{12} \\ \frac{-1}{6} & \frac{4}{9} \end{pmatrix}$. Find the value of x?

a. $\frac{28}{3}$ b. $\frac{32}{3}$ c. $\frac{34}{3}$ d. 10

(22) According to the laws of propositional logic,

I. $p \wedge true = p$
II. $p \vee true = false$
III. $p \wedge false = true$
IV. $p \vee false = p$; Which are the incorrect?

a. I and II only
b. II and III only
c. III and IV only
d. I and IV only

(23) If $(2x \quad 3)\begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix}\begin{pmatrix} x \\ 8 \end{pmatrix} = 0, and\ x > 0$ then the value of x is,

a. $\frac{-23}{4}$ b. $\frac{23}{4}$ c. $\frac{-23}{2}$ d. $\frac{-23}{2}$

(24) Which are the correct statements given below?

I. $\mathbb{Q}$ number set is a subset of $\mathbb{Z}$.
II. Any $\mathbb{R}$ number is a $\mathbb{C}$ number.
III. $\mathbb{N}$ number set is a proper subset of $\mathbb{Z}$.

a. II only b. III only c. II and III d. I and III

(25) What can be said regarding a line if its slope $(tan\theta)$ is negative?

a. $\theta < 90^0$
b. $\theta > 90^0$
c. It is parallel to the x- axis
d. None of all

(26) The equation of straight line passing through the point (1, 2) and parallel to the line $y = 3x + 1$,

a. $y + 2 = x + 1$ b. $y + 2 = 3(x + 1)$ c. $y + 2 = 3(x - 1)$ d. $y - 2 = x - 1$

(27) A proposition that simply means exactly one of p and q will be true but both cannot be true is called as,

a. Tautology b. Contradiction c. Exclusive or d. None of all

(28) The value of determinant given below: $\begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ -(b+c-a) & -(c+a-b) & -(a+b-c) \end{vmatrix}$

a. 0 b. ab + bc + ca c. abc d. a + b + c

(29) The subset of $\mathbb{N}$ are shown in;

a. $\{-1,0,1\}$ b. $\{0,1,2,3\}$ c.$\{-\frac{1}{2},0,\frac{1}{2}\}$ d. {2, 3, 4}

(30) $A = \begin{pmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{pmatrix}$ *and determinant of* $A = 0$. Value of x ?

a. 4 b. 6 c. 8 d. 12

(31) Identify which sets are equal.

$A = \{ X \mid X \text{ is an integer where } X > 0 \text{ and } X < 6\}$

$B = \{ X \mid X \text{ is an integer whose square is } > 0 \text{ and } < 36\}$

$C = \{X \mid X \text{ is a positive integer whose square is } > 0 \text{ and } \leq 36\}$

a. A = B b. B = C c. A = C d. None of all

(32) The equation of the line which cuts off equal and positive intercepts from the axes and passes through the point (α, β) is,

a. $x + y = \alpha + \beta$ b. $x + y = \alpha$ c. $x + y = \beta$ d. None of all

(33) How many prime numbers are there that are less than 50?

a. 16 b. 15 c. 14 d. 13

(34) What is logically equivalent to the following statements?(Note that fail is equivalent to not pass) p: "$I \text{ pass only if you pass}$"

I. You pass only if I pass
II. If you fail then I fail
III. If you pass then I pass
IV. You fail if I pass

a. I only
b. II only
c. III only
d. IV only

(35) A circle always passes through the fixed point (a, 0) and (-a, 0). If the given points are the end of a diameter, then the equation of a circle,

a. $x^2 + y^2 = a^2$
b. $x^2 + y^2 + a^2 = 0$
c. $x^2 + y^2 + 2x + 2y = a^2$
d. $x^2 + y^2 - 2x - 2y = a^2$

(36) Simplify the following expression. $(2a)^2(-2a^3b^{-4})(ab^7)$

a. $8a^5b^3$ *b.* $-8a^6b^3$ *c.* $-4a^6b^3$ *d.* $4a^5b^3$

(37) Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ *and* $B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Then $|A + B|$?

a. 2 b. 4 c. 6 d. 8

(38) Find the simplified form of the expression that need to have positive exponents. $\frac{2^8p^{-6}q^4}{(2pq)^3}$

a. $\frac{32q}{p^3}$ b. $\frac{-32q}{p^3}$ c. $\frac{-32q^3}{p^2}$ d. $\frac{32q}{p^9}$

(39) Find the equation of a circle, if the end point of the diameters are (3, 2) and (2, 5).

a. $x^2 + y^2 - 5x - 7y - 16 = 0$
b. $x^2 + y^2 + 5x + 7y - 16 = 0$

c. $x^2 + y^2 - 5x - 7y + 16 = 0$

d. $x^2 + y^2 + 5x - 7y + 16 = 0$

(40) Let $B = \{ X \mid X = n^2 + 2 \; such \; that \; -2 \leq n < 2\}$. What is the $|B|$?

a. 2 b. 3 c. 4 d. 5

(41) Let $A = \begin{pmatrix} -2 & 5 \\ 1 & -1 \end{pmatrix} and \; B = \begin{pmatrix} 2 & 6 \\ 3 & 2 \end{pmatrix}$. Then $A^{-1} + B^{-1}$?

a. $\frac{1}{42}\begin{pmatrix} -8 & 88 \\ 23 & -22 \end{pmatrix}$ b. $\frac{1}{42}\begin{pmatrix} 8 & 88 \\ 23 & 22 \end{pmatrix}$ c. $-\frac{1}{42}\begin{pmatrix} 8 & -88 \\ -23 & 22 \end{pmatrix}$ d. None of all

(42) Let $p^{-x}q^{y} = \frac{q^{-1}(\sqrt[3]{p})}{(\sqrt[4]{p})q^{-8}}$. What are the values that represented by *x* and *y*?

a. $\frac{-1}{12}, 7$ b. $\frac{1}{12}, 7$ c. $\frac{1}{12}, -9$ d. None of all

(43) A determinant of a 3 x 3 matrix is equal to 15. If the second and the third columns are interchanged, is the new determinant value,

a. 15 b. -15 c. 45 d. -45

(44) Oshan said that $\sim [(p \vee q) \wedge r)] = (\sim p \wedge \sim q) \vee \sim r$.
Nadun said that $\sim (p \wedge \sim q) = \sim p \vee q$. Who said the truth?

a. Oshan b. Nadun c. Both d. None

(45) Find the center and the radius of the circle. $(x+2)^2 + \left(y - \frac{1}{2}\right)^2 = 1$

a. $C \equiv \left(-2, \frac{1}{2}\right), r = 16$

b. $C \equiv \left(2, \frac{1}{2}\right), r = 1$

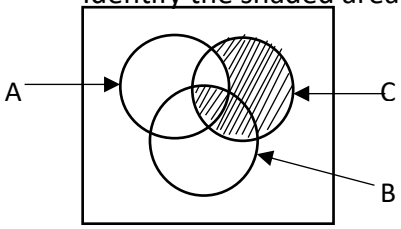
c. $C \equiv \left(-2, \frac{1}{2}\right), r = 1$

d. $C \equiv \left(-2, \frac{-1}{2}\right), r = 1$

(46) Let $A = \{1, 2, \{3,4\}, 5,6\}$ and $B = \{3, 4\}$. Arvin said that $B \subseteq A$. Sisila said that B is an element of A. Who said the truth?

a. Arvin b. Sisila c. Both d. None

(47) Identify the shaded area in the given Venn diagram.



a. $(A/B) \cup C$
b. $(A \; / \; B^{-1})^{-1} \cap C$
c. $(A \; / \; B)^{-1} \cap C$
d. None of all

(48) p: $Indian \; army \; moves \; back$, q: $Chinese \; army \; moves \; back$, r: $There \; is \; a \; war$.
Then what is the statement that "if Indian army moves back and Chinese army moves back, then there is no war"

a. $(p \vee q) => r$ b. $(p \wedge q) => r$ c. $(p \wedge q) =>\sim r$ d. None of all

(49) $[(p <=> q) \wedge (q => r)] => (p => r)$ this proposition shows a,

a. Tautology b. Contradiction c. Contingent d. None of all

(50) Identify the correct statements.

I. $A \cup \phi = \phi$
II. $A \cap \Omega = A$
III. $A \cup A^{-1} = \phi$

a. I only
b. II only
c. III only
d. II and III only

Question 2 **[25 marks]**

(1) Simplify the following simultaneous equations by using the matrix form.
$2x + 5y - 9 = 0,\ 4y - 3x + 7 = 0$ [5 marks]

(2) $A = \begin{pmatrix} 3 & -2 & 5 \\ 7 & 9 & -6 \end{pmatrix}, and\ B = \begin{pmatrix} -8 & 4 & -12 \\ 16 & 8 & -8 \end{pmatrix}$. If $2A + kB = \begin{pmatrix} 8 & -5 & 13 \\ 10 & 16 & -10 \end{pmatrix}$, find the k? [5 marks]

(3) Find the determinant of the following matrix. $D = \begin{pmatrix} 7 & -8 & 5 \\ -3 & 6 & 4 \\ -9 & -4 & 3 \end{pmatrix}$ [5 marks]

(4) Let $2x + 3y - 3z = 12, 9x + 6y = 10, and\ 8x + 5y - 5z = 11$. Find the x, y, and z values by using **equation solving techniques of matrices**. [10 marks]

Question 3 **[25 marks]**

(1) Find the equation of a line (L1) through point (-2, -5) with gradient $\frac{-3}{7}$. Then write the equation of a line (L2) which perpendicular to L1 and containing point $\left(\frac{-1}{3}, \frac{2}{3}\right)$. [5 marks]

(2) Let line $L3 \equiv 2x - 3y = 5\ and\ L4 \equiv -3y + 5x = -2$ intersect at point P. What is the coordinate of P? [5 marks]

(3) What is the perpendicular distance from $L5 \equiv 2x + 8y = 11$ to $B \equiv (-2, -3)$. [5 marks]

(4) Two gun shooters located in the positions which given below.
$P_1 \equiv \big(2 + 3cos(30), -3 + 3sin(30)\big), P_2 \equiv \big(2 + 3cos(60), -3 + 3sin(60)\big)$ Identify the straight line equation and circle equation which go through this points. [10 marks]